

Softly Non-Spoken: Exploring the Potential of Soft Robotics to Support Non-Verbal Communication

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ABSTRACT

Existing Augmentative and Alternative Communication (AAC) technologies aim to support people with communication disability participate more fully in life and society, but these often prioritize linguistic and spoken modes of expression. Embracing a broader view of communication that includes non-verbal and embodied expressions, this interdisciplinary project explored the potential of soft robotics for developing AAC technologies that create opportunities for more nuanced forms of communication. To do this, the Softly Non-Spoken project facilitated interdisciplinary workshops and developed soft robotic objects which were then used as probes to explore how such technologies might support expressive, affective non-verbal communication. Results suggest that soft robotics materials and functions offer potential in this design space, including compelling possibilities for object-enabled human interaction that could offer more individual control around communication. The findings also propose a conceptualization of communication that encompasses embodied expression and could cultivate ambiguity and intrigue in future AAC design.

CCS CONCEPTS

• **Human-centred computing**; • **Human computer interaction (HCI)**: **Accessibility**;

KEYWORDS

Augmentative and Alternative Communication, Communication, Disability, Soft Robotics

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1 INTRODUCTION

Around 20% of the population of the UK will experience communication disability across their lifespan [1]. People experience communication disability for a range of reasons; some acquire communication disability at birth or in early childhood due to conditions such as cerebral palsy, while others develop disabilities in later life because of acquired conditions such as stroke, cancer, or neurological condition. Communication has been recognized as essential to societal participation [2] and communication impairments can negatively impact aspects of life, including educational outcomes, employment opportunities [3], and social engagement [4]. Augmentative and alternative communication (AAC) refers to strategies that can be used by people who experience communication disability to support them to communicate [5]. AAC include no- and low-tech strategies like gestural signs and pictures that people point to and also high-tech, powered devices that can produce synthetic speech by activating messages that are entered in to or pre-stored within them [6].

AAC devices have traditionally been electronic and have more recently tended to be computer-based. They are often screens or boxes that can be mounted onto wheelchairs and consist of an input access mechanism (a keyboard, picture grid or switch) and an output (synthetic voice, screen or both). AAC device design has historically focused on creating a technological replacement for speech and language. Inputs are semantic representations, that is they are either text-based (where a person inputs words and sentences using a keyboard) or consist of pictures or symbols that represent words or phrases (where the user selects pictures or symbols from a grid of pre-programmed options). The limitations of current approaches to AAC device design result in the low uptake of devices that could serve to improve the health and wellbeing of people who have communication disability. Current AAC devices have been shown to be unreliable and insufficiently durable [7] and costly or difficult to access [8]. Successful AAC is also contingent on the skills and abilities of the individuals who use the device [9]. Within the HCI accessibility community, challenges and shortcomings have been identified in development and use of AAC technologies, including high rates of abandonment, lack of reliability, and poor consideration of how they might be used [10]. Furthermore, AAC users can find it frustrating when they are unable to emotionally connect with other people because they

cannot express themselves or their sense of identity sufficiently through AAC-mediated communication [11].

Typical and widely used AAC systems are based on a conceptualisation of communication that is rooted in speech, language, and the transmission of messages from one person to another [12]. As such, many digital AAC technologies prioritise verbal communication and ‘voice’ [13] through recreating spoken messages or content for the person using the device [14]. However, the concept of ‘communication’ has also been more broadly understood to go beyond a transactional exchange of information and include multi-modal, non-verbal or embodied means of connection between people, allowing them to engage in spontaneous, socially mediated interactions [15]. While those with communication difficulties can be adept at finding multiple and multi-modal ways to communicate themselves, such aspects of embodied communication are often overlooked in current forms of AAC. Furthermore, many AAC technologies designed to support disabled people are developed alongside a medicalized perspective of disability, in which the technology is seen to ‘fix’ an impairment or help the person to become more ‘normal’ [16]. Such interventions may also not account for the importance of aspects like emotional health and environmental circumstances (e.g., social support) as fundamental to quality of life for people who have communication disability [17]. AAC device use is deeply connected to the social and cultural context in which it is used, but that context is often overlooked in the design, use and study of the devices [13]. There is, therefore, a need for AAC devices that conceptualise ‘communication’ beyond verbal, speech-oriented interaction and consider context of use. In this paper, we present preliminary work that explores how design of AAC technologies could support broader approaches to communication, including non-verbal and embodied forms. We are specifically interested in the potential that soft robotic technologies offer in this arena, an area that is currently underdeveloped. To this end, we held 4 interdisciplinary workshops with professionals and 3 public stakeholder events that examined responses to soft robotic design probes in relation to non-verbal communication. We present design insights that could facilitate future development of soft robotics technologies that enable broader forms of communication.

2 BACKGROUND AND EXISTING WORK

2.1 Design to support non-verbal, embodied communication

Previous Human-Computer Interaction research has encouraged a shift in AAC technology design towards a more embodied and contextualised conceptualisation of communication. For example, Valencia et al [18] examined the role of motion-based AAC and physically expressive objects. Schultz Ascari et al [19] demonstrated the importance of non-verbal and embodied communication by using computer vision and machine learning to enable personalised interaction with gestures, and Neate et al [20] explored expression through technology-enabled creativity. Conceptualising communication and expression in these ways shifts the emphasis away from specifics of speech and language towards a wider set of dynamics of human connection. Arguably, existing AAC technologies that focus on replicating or mimicking spoken communication will not support people to access the range of communicative modes that

enable social closeness. Nor do speech and language alone allow for the expression of individuality, responsiveness, and subtlety that enable humanness and facilitate interpersonal connections. For example, during conversation, body language is often used to indicate agreement or disagreement or a desire to interject. This project aims to pursue this challenge of designing and developing AAC technologies that are capable of and attuned to more embodied, non-verbal, and expressive modes of communication.

2.2 The Potential of Soft Robotics for AAC

Soft robots are a subclass of robots that comprise soft and compliant materials which have mechanical properties similar to biological tissues. As a result, soft robotic actuators are increasingly being used where organic motion is desired, for example to generate biomimetic motion or adapt passively to the surrounding environment for reasons such as safety [21]. This includes instances where non-verbal communication is the objective, for example humanoid facial expression generation [22]. As such, the field of soft robotics offers a rich area to explore in examining how technologies and motion may enable or support non-verbal, embodied and expressive forms of communication. The field of robotics has also contributed to an understanding of how humans achieve connections that extend beyond words and speech through exploring aspects of embodied rather than linguistic communication [23]. However, little research has been done on how soft robotics might support those with communication disability and a better understanding is needed on how humans interact with soft robots [24] and the ethical implications of this interaction for design [25, 26]. Existing work involving all types of robots and those with impairments often focuses on the use of robotics for therapy, interventions related to social and cognitive development, or rehabilitation purposes [27, 28; e.g., mobile robot Teo [29]; the huggable [30]. This project aimed to build on these existing studies and explore soft robotic possibilities in particular that extend beyond therapeutic interventions into supporting and enabling expressiveness, agency, and relationality in communication.

3 METHODOLOGY

Formed as an experimental partnership in 2021, Softly Non-Spoken is an interdisciplinary project that aims to explore the potential for soft robotics in broadening the current range of AAC technologies alongside a conceptualization of communication that considers expression, nuance and non-verbal strategies. This exploratory collaboration was a small-scale funded project that brought together a team of roboticists, puppeteers, speech and language therapists, and education researchers who shared an interest in how the design and use of technologies could support communication beyond therapeutic functions. In this paper, we report on two areas of initial design activity: 1) a set of knowledge exchange and design workshops and 2) design probe engagement with public stakeholder groups. The project activity began with interdisciplinary knowledge exchange workshops involving project members and aimed to first construct a shared understanding of communication and existing AAC user contexts. These workshops involved the four primary team members (an academic from robotics and education fields, a speech and language therapist and two roboticists from a

private company), as well as an MSc robotics student and research associates from the Bristol Robotics Lab. These involved shared reading (Valencia [18], Wachsmuth et al [23]) and interactive activities. The interdisciplinary workshop aims and activities were as follows:

- **Workshop 1:** This workshop aimed to develop shared understanding across interprofessional practice and disciplines of speech and language therapy, disability and soft robotics, including the affordances of these technologies for communication. It involved creative activities exploring the role of objects as mediators of communication, existing AAC devices, and potential affordances of soft robotics in this space.
- **Workshop 2:** This workshop introduced low-fi soft robotics technologies and examples alongside discussion of core texts to co-create a mutual understanding of ‘communication’ and develop design challenge areas.
- **Workshop 3:** Project members participated in a puppetry workshop to consider how users experience external objects. The use of puppetry helped us consider how soft robotics materials and features promote expression.
- **Workshop 4:** A final team workshop involved interacting with the soft robotics probes, with ongoing iterative refinement alongside project aims and consideration of communication.

We further attended three public engagement events (1 workshop at an academic AAC conference involving AAC users, educators and carers) and attendance at 2 public-facing ‘FUTURES’ research events, one for schools and one for general public) with the design probes to seek initial engagement with and feedback on their use and value in supporting non-verbal, embodied communication. At these events, we shared information about the project, invited interaction with the objects described above and gathered responses to them. The events were aimed at different audiences and so offered a range of responses to the project and soft robotic objects. In all, more than 120 people informally interacted with the project and objects at these events. In each of the sessions, there were two primary activities, adapted according to the event context. The first activity asked people to select from a series of displayed material objects that they felt they could use to communicate and then to describe how they would use that for communication. The intention of this activity was to provoke audiences into expanding ideas about different ways to communicate and the role of objects in this interaction. We then briefly explained the project and showed the different soft robotics technology probes we developed to observe their interactions and gather responses to these through written and verbal comments.

4 RESULTS

The results of the project outlined below offer insight for future research into AAC device design with soft robotics by demonstrating a) the value of inter-professional working to add co-created conceptual understanding and depth to design methods and b) the potential for soft robotics to support expressive, non-verbal communication.

4.1 Redefining and designing for ‘communication’ in non-verbal and embodied ways

As an interdisciplinary project developed in response to a recognized limitation in how ‘communication’ is conceptualized in current AAC devices, we prioritized developing a shared understanding of project concepts before commencing design work. Therefore, initial workshops prioritized co-creating mutual understanding of ‘communication,’ as a *multi-modal, interactional and interpretative process of meaning making that is dynamic, relational and uncertain*. [12]. Using this definition, we developed a set of design challenge areas to concentrate on in this project, as discussed next.

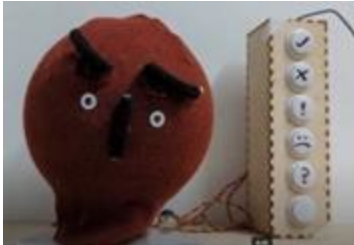
4.2 Considering attention, empathy and provocation in AAC design

Through the workshops, we developed three design challenge areas to focus on. These responded to limitations of existing AAC systems and focused on conceptual areas in need of consideration in AAC design. The design challenge areas were:

1. **Attention:** related to the role of attention and attentiveness in communication (How might AAC technologies demonstrate or attract attention in communication or interaction?)
2. **Empathy:** related to the potential of communication to enable human connection and understanding (How might AAC systems support human connection and understanding of others?)
3. **Provocation:** related to the possibilities of communication as spontaneous, surprising and intriguing (How might AAC systems provoke interaction or allow for spontaneous or unscripted communication?)

Inter-team discussions about these design challenges led to development of use case scenarios and initial soft robotics models that responded to the challenge areas. Development of initial models also used insight from the preceding workshops, including, for example, qualities of ‘embodied communication,’ like the role of facial features in shaping shared attention or expressing patency or how the size and positioning of external communication devices could influence the behaviour of potential communication partners. These insights contributed to the development of three soft robotics objects (see Table 1) that were developed to share and gain feedback from stakeholders. These soft robotics objects were intended to be used as design probes, in that they were ‘both tools for design and tools for exploration.’ [31]. Across their varied history in design and HCI fields, design or cultural probes have been used to enable and prompt engagement and responses related to certain contexts and experiences [32, 33]. In this project, we aligned with Wallace et al [31], who described technology design probes as small-scale, ‘directed craft objects.’ Probes have also been used as ways to focus on emotion in interaction design and offer opportunity to engage with more nuanced and immaterial ideas of non-verbal communication, emotion and expression [33] and to bring qualities of openness and playfulness to early design processes, all of which aligned with this project’s aims. Thus, the aim of using these artefacts as probes was two-fold: to trigger interaction and communication through their use and to also prompt feedback and reflections on communication

Table 1: Brief descriptions of the soft robotic objects developed in the project

Object	Image	Description	Communication potential
Fan		Fan-based objects that could expand, contract, deflate or use wave-like motion. These were made using different material and some also held written messages that would be shown on expansion.	Possibility of signalling attention, lack of interest, or emotional responses. Developed in response to the design challenge of ‘Attention.’
Silicone Actuator		A set of silicone actuators pneumatically powered via a syringe to expand or curl, deflate, and contract. Texture and colour were added with coffee grounds and small plastic pellets.	Exploring multi-sensory, tactile objects as provocateurs for reaction and interaction. This object was intentionally ambiguous and provided more nuanced communicative intent. Developed in response to the design challenge of ‘Provocation.’
Expressive head		This object was constructed of foam and textiles. Its expression changed through moving facial features and a moving head, which was controlled through pushing buttons that signalled certain intended expressions.	Explicit and familiar expressions to communicate single messages non-verbally. Developed in response to the design challenge of ‘Empathy.’

from those using them. Designed with this purpose in mind, the objects were intended to support ‘open-ended, provocative and oblique’ engagement activities to aid and further our own understanding of object-mediated communication, contexts of use and participant experiences of soft robotic-oriented AAC [33]. They were developed alongside consideration of the design challenge areas, as shown and described in Table 1 below.

4.3 Soft robotics’ potential in developing expressive and non-verbal forms of AAC technologies

Results from the public engagement events supported the premise of this project that suggested further development of AAC is needed, particularly in relation to widening the focus beyond verbal and voice-oriented strategies. For example, at the AAC conference, stakeholders recognised the limitations of existing AAC strategies and expressed a need for mechanisms allowing AAC users to be more expressive and have more control over communication and dialogue. This was discussed in relation to existing AAC use issues,

for example, the length of time it can take to construct a message on AAC devices and the need for holding space or attention for AAC users while they create those messages. During the public research event, those interacting with the project information and probes shared observations about the potential for objects to mediate interactions where words and spoken language could be difficult or challenging. Responses to the design probes themselves provided unique insight into what the different material, features and functions of the artefacts offered. The expressive robotic head generated significant attention and stakeholders found the texture and movement of the robotic head either amusing or alarming. The facial features unnerved some, raising questions about the value of humanoid robots as interaction partners. This object was also primarily seen as capable of expressing certain explicit messages like ‘yes’ or ‘no’ or emotions like ‘angry,’ resulting in interaction related to transmission of ‘correct’ messages. Thus, message reception was the primary way of interacting with this object, with participants wondering if they were interpreting the expression correctly. In contrast, interaction with the fans was more emergent and diversified; they were used to demonstrate attentiveness (i.e. standing up

straight), different moods or emotions (i.e., bowed down if ‘sad’), playfulness (i.e., becoming wings to fly away), though some stakeholders also suggested they would find it difficult to ‘communicate’ with the fan as it was seen to have less potential to generate clear messages. Attendees were more likely to manipulate, touch, or squish the pliable silicone actuators but this material also generated the strongest tactile responses, enacting a stronger reaction one way or another (attraction or repulsion). The silicone actuator and fan were seen to be more ambiguous in their communication potential, but these were also used, responded to or manipulated in more creative ways. Furthermore, these tactile interactions were controlled and managed by individuals and the actuators also mediated and provoked interaction between participants that went beyond a focus on the technology itself.

These engagement activities suggest several insights in response to the project aims and in relation to the potential of soft robotics to support various ways of communicating. The design and sharing of the probes suggest that the motion afforded by soft robotics can lend themselves well to more expressive and non-verbal communication. The fan objects were useful in gaining attention from others and then enabling interaction between people. Soft silicone material invited touch and haptic, responsive interaction in ways that more rigid materials do not which in turn elicited provocation through intrigue, surprise and creative interpretations of the possible messages or signals they implied. Notably, many of the interactions began from an initial assumption of ‘communication’ as related to delivery of messages via speech and voices. However, through further interaction with the objects, more expressive, nuanced and non-verbal forms of communication emerged. In this way, interaction with the objects themselves facilitated a reconsideration of what ‘communication’ involves. This short-term project did not intend to use this stakeholder engagement as ways to gather ‘data’ to generate objective, validated knowledge about AAC use and soft robots but was intentionally opened to allow for nuance, indeterminacy and expression [34]. Thus, these objects enabled us to consider what ‘communication’ is and can be, in what instances non-verbal forms of expression might be valued or utilized, and how the affordances and qualities of soft robotics could support expression for people who have communication disability.

5 DISCUSSION

Through its collaborative, interdisciplinary approach to design, this small-scale project has developed early insights on the ways in which soft robotics could support more holistic communication for those who use AAC. In this section, we specifically summarise the potential for soft robotics in supporting non-verbal communication, as well as broader insights related to conceptualisations of ‘communication’ in AAC and interdisciplinary working.

5.1 Potential of Soft Robotics to Support Non-Verbal Communication

The exploratory work carried out during this project suggests there is significant potential for soft robotics technologies to support communication beyond verbal transmission of messages through their facilitation of object-enabled human interaction and allowance of

individual control in communication processes. The term ‘human-ness’ has been attributed as one of the desired outcomes from using AAC [11, 35] and captures the power of AAC in relation to an individual’s ability to realise their whole self and maintain autonomy. Independence, agency, and self-identity are important elements of wellbeing and quality of life for people who have communication disability, along with social closeness [17]. While having access to a synthetic voice can provide a form of self-expression, how devices are used within a social context is more indicative of a sense of identity [36]. Current AAC designs can promote an individual’s independence and agency by enabling speech and language functions but tend to be less successful at enabling expression of individual identity or supporting the more implicit signaling that often enables social connectivity and leads to social closeness [14].

Findings from this project suggest that soft robotics offer opportunities for different forms of object-enabled communication than are common in current AAC devices. Responses to the objects indicate that soft robotics offer opportunities for interactions between people and with objects that current screen-based and AAC technologies may not. In this sense, they can move beyond single user-device interactions and turn attention to interpersonal relationality. Furthermore, the soft robotic objects demonstrated ways to support non-verbal expression that could augment and be used alongside existing AAC technologies - for example: supporting turn-taking; signalling being open or closed to communication; managing pace and timings of communication; and opportunities for play and humour. The tactile nature of the objects afforded opportunities for people to manipulate and control responses that current pre-programmed AAC devices do not. This supports another important consideration in AAC design: providing the person using the technologies with control and agency around what, when, and how they communicate. Finally, as a work-in-progress, this project offers insight into how people interact with and respond to soft robots and materials, a need identified by Jorgenson et al [24]. In this vein, the workshops expanded our understanding of possible use cases, seeing the potential for these technologies to support non-verbal expression beyond typical AAC users or those with communication impairments – for example, for children in classrooms who may struggle with speaking out.

5.2 Re-Conceptualising ‘Communication’ in AAC and with Objects

Through the use of soft robotics design probes, this project also explored conceptualisations of communication and how objects (and AAC) enable interaction between people. This can be a provocative, challenging space, particularly in considering ‘communication’ as an embodied form of interaction, as suggested by the propensity of stakeholders defaulting to verbal or spoken language when asked to think about how objects help us communicate. The stakeholder workshops suggested that some objects invite interaction between people, rather than accommodating interaction with the device itself (as tablet-based AAC might). Objects can also be used to show non-verbal messages or provide opportunities to opt out of communication (e.g., by ‘shutting down’ the fans). Our project suggests that soft robotics have potential to enact a broader notion of communication that sees the importance of embodiment and

ambiguity in human connection [12]. This may be a provocative design space for AAC but a vitally important one in supporting wholeness in human communication.

Additionally, the project presented a unique opportunity to exploit the potential of soft robotics to explore ways to better understand communication through movement, gesture, or touch, which open opportunity to explore more autonomous, unscripted ways of interacting. Sociological conceptualisations of communication allow us to consider the relational, interactional, and uncertain qualities of human connection [12, 15, 37]. They also provide professionals and researchers with the philosophical foundations to shift away from the verbal or linguistic aspects of communication that have traditionally informed AAC device design and foster greater attention to the human dynamics of communication. Communication is more than getting a message across and extends beyond a functional purpose. Considering the embodied, multi-modal forms of expression can enable social closeness and the expression of human identity. These more implicit, embodied communicative signals can also be impaired in conditions that cause communication disability and novel non-verbal AAC may form part of the multi-modal system that people use to compensate for such impairments. The object-probes that were not explicitly evocative of a human-oriented message provoked more quizzical, playful and ambiguous interactions with audiences, and were not always associated with typical notions of ‘communication.’ While these objects enabled expressiveness and shared attention between participants, it may be challenging for those who use AAC devices to value what they offer. Thus, reconceptualizing communication as expressive, embodied, and generative of human connection is required alongside development of technologies that purport to enable just that.

5.3 Interdisciplinary Designing

Lastly, the project offers insight into strategies for interdisciplinary work in design. Early on, we explicitly prioritized allocating sufficient time to understanding the different disciplines among the team members and co-developing an understanding of project concepts and aims. Making this stage explicit through creative activities, reading and discussion meant we first developed a shared conceptual grounding before considering robotics design possibilities. This helped us to design objects that reflected project aims across disciplines. It also demonstrated a core benefit in interdisciplinary working of unsettling existing understandings or assumptions within individual fields and allowing ideas to reform in productive ways beyond what a singular discipline might allow. This project intentionally utilized the collaborative, interdisciplinary space to design and develop artefacts with the aim of further provocation. In this sense, the probes were designed with features and functions shaped by a broader definition of communication and with the intention of evoking responses in this vein. It was not intended to ‘understand’ or make broad statements about how a certain community communicates or might use something but rather to gather experiences and co-create understandings that aim to feed into future design responses.

6 CONCLUSION

Findings from the Softly Non-Spoken project offer exploratory insight into what soft robotics technology can offer to future AAC device development. As a project team, we aim to co-develop more technologically capable and robust prototypes that can be iterated collaboratively with people with a communication impairment or who work with AAC. As part of this, we aim to explore how people control soft robotic prototypes, integrate them into existing communication systems, and use them in natural interactions in different contexts and settings. Further considerations in this area include how to consider relevant modalities in soft robotic design for communication and the ethical implications of their use and potential misuse. Additionally, the design and feedback of the soft robotic objects also raised questions about the use of existing AAC in aiding and transmitting explicit messages, questioning the importance of having a message understood accurately in order to communicate effectively, to build relationships and to participate in society. This is another area we hope to explore in future work.

This project and its findings offer a challenge to conventional AAC device design, suggesting potentiality for novel, innovative device design alongside broader conceptualisations of communication. It suggests a need for research and design approaches that examine how AAC technologies and strategies can be developed to support agency and expression in communication and interaction for those who have communication disability, further supporting opportunities for more equal contributions and participation in everyday life. In summary, by pursuing a conceptualization of communication as multi-modal and multi-faceted, this project further sounds a call in the HCI field for development of devices, technologies and systems that go beyond current technical solutions oriented to verbal message transmission.

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